21-242 Recitation 12

Linear Transformations — Q4. Coordinates [ ]

Course-level solution, with a later-course alternative when useful

October 1, 2026

Prompt

Define $[\vec{v}]_{\mathcal{B}}$ (where $\mathcal{B}$ is a basis for V) and prove that $\vec{v} \mapsto [\vec{v}]_{\mathcal{B}}$ is an invertible linear transformation.

Course-level solution

Let $\mathcal{B} = (\vec{b}_1, \ldots, \vec{b}_n)$ be a basis of V. Every $\vec{v} \in V$ may be written uniquely as $\vec{v} = \sum_{i=1}^{n} a_i \vec{b}_i$. Define

$$[\vec{v}]_{\mathcal{B}} := \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} \in \mathbb{F}^n.$$

Let $\Phi_{\mathcal{B}} : V \to \mathbb{F}^n$ by $\Phi_{\mathcal{B}}(\vec{v}) = [\vec{v}]_{\mathcal{B}}$.

Linearity. If $\vec{v} = \sum a_i \vec{b}_i$ and $\vec{w} = \sum c_i \vec{b}_i$, then $r\vec{v} + s\vec{w} = \sum (ra_i + sc_i)\vec{b}_i$, so $[\ r\vec{v} + s\vec{w}\ ]_{\mathcal{B}} = r[\vec{v}]_{\mathcal{B}} + s[\vec{w}]_{\mathcal{B}}$.

Invertibility. $\Phi_{\mathcal{B}}$ is bijective because each coordinate tuple corresponds to exactly one linear combination of $\mathcal{B}$. Explicitly, the inverse is

$$\Phi_{\mathcal{B}}^{-1} : \mathbb{F}^n \to V, \qquad \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} \mapsto \sum_{i=1}^{n} a_i \vec{b}_i,$$

which is linear (it is the unique map sending the standard basis vector e_i to $\vec{b}_i$, by Q3). Thus $\Phi_{\mathcal{B}}$ is an invertible linear transformation, i.e. an isomorphism $V \cong \mathbb{F}^n$.